

Preservation of Causal Versus Correlational Language During LLM-Assisted Editing: Risks, Mechanisms, and Epistemic Integrity

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Abstract

The integration of Large Language Models (LLMs) into the scientific writing process has fundamentally altered how researchers draft, edit, and refine their manuscripts. While LLMs offer significant advantages in fluency, grammar, and stylistic polish, their deployment introduces a critical risk to the epistemic integrity of scientific communication: the inadvertent semantic shift between correlational and causal language. This paper comprehensively examines the preservation of causal versus correlational claims during LLM-assisted editing. We synthesize recent findings demonstrating that LLMs, which operate on probabilistic text generation rather than grounded causal reasoning, frequently distort intended scientific meaning by replacing careful, hedged correlational phrasing with unwarranted causal assertions (or vice versa). By analyzing current literature spanning computational linguistics, epistemology, and biomedical reporting, we identify the technical mechanisms underlying these failures—such as parametric memory override and the lack of robust structural causal models in LLMs. Furthermore, we review current approaches to mitigating these risks, including the development of real-world scientific text datasets and prompt-level constraints. Ultimately, we outline the research challenges, critical limitations of current models, and future directions necessary to ensure that LLM-assisted editing preserves the rigorous causal boundaries essential to scientific discovery.

Keywords: Large Language Models, Scientific Writing, Causal Inference, Epistemic Integrity, Correlational Language, Semantic Shift, AI-Assisted Editing

1 Introduction

The proliferation of Large Language Models (LLMs) has initiated a paradigm shift in academic publishing and scientific communication. Across disciplines, researchers increasingly utilize AI-assisted tools for various stages of the writing process, from initial ideation and drafting to sophisticated developmental editing and proofreading. While the utility of LLMs in enhancing the syntactic fluency and grammatical precision of academic texts is widely acknowledged, their influence on the semantic fidelity—specifically, the precise representation of scientific claims—remains a subject of intense scrutiny (Abdulhai et al., 2026). One of the most critical dimensions of this semantic fidelity is the distinction between causal and correlational language.

In scientific research, particularly within observational studies, biomedicine, and the social sciences, the epistemic boundary between correlation (two variables being associated) and causation (one variable directly influencing another) is foundational. Researchers are trained to deploy specific, often hedged language (e.g., “associated with,” “linked to,” “correlates with”) to accurately reflect the limitations of

observational data, avoiding definitive causal verbs (e.g., “causes,” “determines,” “results in”). However, as LLMs shift from mere post-editing assistants to active co-authors, there is emerging evidence that these models consistently alter the intended meaning of human writing (Abdulhai et al., 2026). When prompted to improve the “flow,” “readability,” or “impact” of a scientific abstract, an LLM might inadvertently upgrade a correlational finding into a causal one, stripping away vital scientific caveats.

This paper investigates the phenomenon of causal language preservation during LLM-assisted editing. We argue that because LLMs lack intrinsic causal reasoning capabilities (Fu et al., 2025) and often fail to accurately infer causal relationships from complex, real-world academic texts (Saklad, 2026), they pose a systemic risk of introducing “spin”—distorted interpretations that mislead readers (Chiu et al., 2017). Through a comprehensive review of the latest developments up to 2026, we explore the theoretical background, technical mechanisms, current mitigation strategies, and future research directions necessary to safeguard the integrity of scientific reporting in the age of AI.

2 Background, Related Work, and Literature Review

2.1 The Epistemology of Causal vs. Correlational Language

The distinction between correlation and causation is not merely a statistical artifact; it is an epistemological cornerstone of the scientific method. Observational studies, which cannot control for all confounding variables, typically yield correlational data. Historically, scientific writing has relied on strict conventions to prevent the overstatement of such findings. However, human readers inherently struggle with this distinction. In a foundational study, Gershman & Ullman (2023) demonstrated that readers frequently draw causal implicatures from purely correlational statements. Even when researchers use orthodox, seemingly non-causal stock phrases (e.g., “X is associated with an increased risk of Y”), laypersons and even domain experts often mentally translate this into “X causes Y.” This innate human bias toward causal narratives makes the precise control of language in scientific manuscripts absolutely paramount; if the text itself introduces causal language where none is warranted, the resultant misinterpretation is magnified.

2.2 ‘Spin’ in Scientific Reporting

The inadvertent or intentional exaggeration of scientific findings is often termed “spin.” Chiu et al. (2017) conducted a methodological systematic review of spin in published biomedical literature, defining it as reporting practices that distort the interpretation of results and mislead readers. They found

that one of the most common manifestations of spin across observational studies and clinical trials is the inappropriate use of causal language to describe mere associations. Spin has severe real-world consequences, potentially misdirecting subsequent clinical research, altering public health policies, and influencing funding allocations based on overstated promises (Chiu et al., 2017). The introduction of LLMs into the drafting process introduces a novel, automated vector for spin, where the exaggeration is not driven by researcher bias, but by algorithmic tendencies toward confident, authoritative-sounding text generation.

2.3 LLMs and the Understanding of Causal Language

Early evaluations of LLMs, such as those conducted on ChatGPT by Kim et al. (2023), revealed significant limitations in the models’ ability to understand causal language in scientific claims. When tasked with labeling the strength of a claim as causal, conditional causal, or correlational, early iterations of ChatGPT lagged considerably behind fine-tuned, domain-specific models. Notably, these models struggled to understand conditional causal claims mitigated by hedges (e.g., “may cause,” “potentially influences”). While subsequent generations of LLMs have grown in parameter size, their fundamental architecture—predicting the next token based on statistical patterns in training data—remains primarily associative rather than causal (Saklad, 2026).

3 Main Thematic and Technical Discussion

3.1 Mechanisms of Semantic Distortion During Editing

When an author prompts an LLM to revise a paragraph containing correlational findings, the model applies learned transformations that optimize for target objectives such as “clarity,” “conciseness,” or “persuasiveness.” Abdulhai et al. (2026) demonstrated that even when LLMs are explicitly prompted to only execute grammar edits based on expert feedback, they still introduce large changes that alter the semantic meaning of the text.

The technical mechanism behind this distortion can be viewed through the lens of what Latif (2026) describes as “Parametric Memory Override” and “Constraint Poisoning.” In high-stakes domain editing, LLMs often ignore explicit user constraints (e.g., “maintain the exact scientific meaning of the associations”) in favor of their parametrically weighted training data. If the model’s training corpus contains a vast amount of text where variables like “smoking” and “lung cancer” are discussed using strong causal verbs, the model will statistically favor those causal verbs during editing, overriding the author’s careful

correlational phrasing. The model conflates statistical likelihood in its training data with scientific truth in the specific context of the manuscript being edited.

3.2 The Absence of Structural Causal Models in LLMs

The failure of LLMs to preserve correlational language is intrinsically linked to their lack of underlying causal reasoning. Fu et al. (2025) systematically analyzed the causal structures of LLMs and Large Reasoning Models (LRMs). By examining Structural Causal Models (SCMs) mapping problem instructions, reasoning steps, and answers, they found that standard LLMs suffer from critical reasoning issues—unfaithfulness and bias—because they rely on superficial correlations. They do not possess a genuine understanding of the physical or logical mechanisms linking two variables. Consequently, when an LLM encounters the phrase “Variable A is linked to Variable B,” it cannot logically assess whether the study design permits a causal claim. It merely acts as a “causal parrot,” generating language that mathematically aligns with the surrounding semantic context, frequently defaulting to more assertive, causal terminology because such language is highly prevalent in declarative texts.

3.3 Evaluation on Real-World Scientific Texts

The inability to preserve causal boundaries is exacerbated by the complexity of real-world academic literature. Saklad (2026) developed a benchmark to test whether LLMs can infer causal relationships from actual scientific texts, which feature varying levels of explicitness, multiple interacting causal events, and complex domains. Their findings indicate that LLMs face significant, perhaps structural, challenges in inferring causal relationships from real-world text, with top models achieving an average F1 score of only 0.535 on such tasks. If an LLM cannot accurately map the causal architecture of a text during the comprehension phase, it is mathematically incapable of reliably preserving that architecture during the generative revision phase.

4 Current Approaches and Developments

4.1 Human-in-the-Loop (HITL) and Constraint Architectures

To mitigate the risks of semantic shift, current developments in LLM-assisted editing emphasize rigid structural controls rather than simple natural-language prompting. Latif (2026) proposes a dual-pass Human-in-the-Loop (HITL) architecture designed to counteract neural network behaviors at the token-generation level. Their research suggests that treating LLMs as autonomous “Auto-Editors” is dangerous

for epistemic integrity. Instead, LLMs must function as “AI Surgical Assistants” operating under the strict, continuous direction of a human domain expert. By utilizing Subject Matter Expert Override Protocols, researchers can lock specific phrases or enforce negative constraints on causal verbs, ensuring the preservation of correlational nuance.

4.2 Dataset Generation for Revisional Modeling

Studying how AI distorts text requires comprehensive baselines of how human scientists actually revise their work. Jourdan et al. (2026) introduced *EarlySciRev*, a dataset of early-stage scientific text revisions extracted from arXiv LaTeX source files. By aligning commented-out discarded text with final published text, researchers have created over 578,000 validated revision pairs grounded in authentic human drafting traces. Datasets like *EarlySciRev* are crucial for the current development pipeline. They allow researchers to evaluate LLM-assisted editing by comparing the model’s suggested revisions against actual human revisions, specifically measuring the rate at which human authors preserve, hedge, or alter causal claims compared to their algorithmic counterparts.

4.3 Advancements in Training Techniques (RLVR)

At the model training level, there are ongoing efforts to instill genuine causal structures into LLM reasoning processes. Fu et al. (2025) observed that models trained with advanced reinforcement learning techniques (such as RLVR) exhibit enhanced causal reasoning capabilities. These training methods help reduce the model’s reliance on spurious correlations and strengthen genuine causal patterns. While these techniques are currently primarily applied to logical reasoning and math tasks, their integration into text-editing models represents a promising avenue for reducing unwarranted causal shifts in generated text.

5 Research Challenges and Limitations

5.1 The Subjectivity of Linguistic Interpretation

A primary challenge in preserving causal versus correlational language is the inherent subjectivity and ambiguity of human language itself. As noted by Kim et al. (2023), choosing appropriate causal language that accurately reflects the strength of evidence is a non-trivial task even for human scientists. Evaluating whether an LLM has inappropriately altered a claim requires robust, objective annotation guidelines, which are notoriously difficult to standardize. The boundary between a “strong correlation”

and a “weak causal link” is often debated within specific scientific sub-domains, making algorithmic alignment incredibly challenging.

5.2 Affective and Psychological Impacts on Authors

The deployment of LLMs as co-authors also introduces psychological limitations that compound the issue of semantic distortion. Zhang (2026) investigated AI-associated academic writing anxiety among postgraduate students, noting that the automation of linguistic execution introduces new evaluative tensions and threats to authorship integrity. When human authors rely heavily on LLMs, they often experience a phenomenon described as “Reactive Writing” (Abdulhai et al., 2026; Zhang, 2026), wherein the author’s primary role shifts from active ideation to passive evaluation of AI suggestions. Under the cognitive load of evaluating rapid AI outputs, authors—particularly novice researchers or non-native speakers—may overlook subtle semantic shifts, inadvertently accepting a causal verb that overstates their correlational data.

5.3 Lack of Established Benchmarks for Tone and Epistemic Shift

While there are robust benchmarks for factual hallucination and grammatical accuracy in LLMs, there is a distinct lack of standardized, automated metrics for evaluating “epistemic shift.” Current NLP evaluation tools struggle to quantify the precise degree to which an edited text has drifted along the correlation-causation continuum. Without large-scale metrics, the scientific community must rely on manual, time-consuming peer review to catch AI-induced spin, a process that is already severely strained.

6 Research Gaps and Future Directions

6.1 Developing Epistemic-Preserving LLMs

Future research must prioritize the development of “Epistemic-Preserving” editing models. This involves fine-tuning LLMs specifically on datasets where edits are strictly constrained by the causal architecture of the original input. Future iterations of models need explicit training on the principles of scientific reporting guidelines (e.g., CONSORT, STROBE) to inherently recognize when variables in an observational study are being discussed and automatically restrict the generative vocabulary to correlational terminology.

6.2 Automated Spin-Detection Agents

There is a critical gap in post-editing verification tools. Future developments should focus on creating specialized, lightweight classifier models—functioning as automated peer reviewers—that analyze differences between original drafts and LLM-edited drafts. These models would highlight any escalation in causal strength, flagging terms like “determines” or “causes” if the underlying methodology identified in the text is purely observational. Integrating such tools into common word processors or LaTeX editors would provide a necessary safeguard against passive semantic shift.

6.3 Longitudinal Studies on Scientific Literature

As AI-assisted writing becomes ubiquitous, longitudinal scientometric studies are required to track shifts in the aggregate causal language of published literature. Future research should mine databases like PubMed and arXiv to determine if the post-2023 era exhibits a statistically significant increase in causal assertions within observational studies. Such empirical evidence will be vital for updating editorial policies at major academic journals.

7 Conclusion

The integration of Large Language Models into scientific writing offers unprecedented efficiency but carries profound risks for the epistemic integrity of scientific communication. This paper has detailed the mechanisms by which LLMs—operating as probabilistic engines devoid of genuine causal reasoning—frequently fail to preserve the crucial distinction between correlational and causal language during the editing process. This semantic distortion acts as an automated form of “spin,” exaggerating scientific claims and potentially misleading readers, policymakers, and subsequent research efforts.

Current models exhibit significant limitations, including parametric memory overrides and an inability to map complex causal relationships in real-world academic texts. While approaches such as Human-in-the-Loop constraints and enhanced reinforcement learning techniques show promise, they require rigorous implementation by vigilant human domain experts. As the scientific community navigates this transition, it is imperative to develop specialized, epistemic-preserving editing models and robust automated spin-detection tools. Ultimately, researchers must recognize that while LLMs can synthesize syntax, the responsibility for scientific causality—and the careful, precise language required to convey it—remains an exclusively human endeavor.

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